Dr. Stéphane Gaël R. Ekodeck
University of Yaounde I, LIRIMA, Cameroon
UMMISCO, IRD France Nord, France
Sorbonne Université, France
stephane-gael.ekodeck@facsciences-uy1.cm

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Professor Tiziano Bianchi
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Dear Professor Bianchi,

We submit the revised manuscript JVCI-26-1814, "Net-Aware Learned Block Mapping for Reversible Data Hiding in Binary Images," for further consideration in the *Journal of Visual Communication and Image Representation*. We thank you and the reviewers for the detailed comments, which led to a substantial revision of the manuscript, experiments, figures, bibliography, and response package.

The revision clarifies the contribution by centering it on serialization-aware activation. The manuscript now states explicitly that net-capacity accounting, binary RDH, PEAK/ZERO block mapping, side information, and distortion-aware ranking are established concepts. The technical contribution is defined as serialization-aware mapping activation: the active reversible PEAK/ZERO configuration is selected according to usable mapping capacity after charging the exact serialized description required for deterministic recovery.

The main changes are as follows:

- The abstract, introduction, related work, discussion, and conclusion were rewritten around this refined contribution.
- An explicit encoder/decoder contract and executable auxiliary-stream model now clarify that the serialized side information is stored or transmitted alongside the marked image.
- A central activation ablation isolates the effect of serialization-aware selection: mean net payload increases from 2139 to 2245 bits relative to an otherwise identical wire-charged activation.
- Additional ablations compare CNN ranking with Hamming, direct DRD, transition-preserving, and connectivity-aware rankers, and a CNN architecture diagnostic now covers depth, window size, width, and inference time.
- The experiments now include serialization sensitivity for baselines, module-level runtime, confidence intervals in the payload/DRD figures, detectability-cause diagnostics, visual residual comparisons, and channel stress tests.
- The claims are narrowed: the method is positioned as capacity-oriented reversible annotation over lossless channels, with detectability-aware interpretation at high payload.
- The related work was shortened and corrected, with additional relevant binary-cover, side-information, and distortion-aware references.

The submission package includes the revised source manuscript and figures, updated highlights, a clean PDF, a PDF with changes shown in blue, a PDF with deletions shown in red and additions shown in blue, and a point-by-point response to every reviewer comment.

Submission statements. The manuscript is original, has not been published previously, is not under consideration elsewhere, and has been approved by all authors. The authors have no known competing financial or personal interests. The reproducibility materials and experiment outputs are described in the manuscript and are publicly available through the GitHub repository cited in the Code Availability statement.

We believe the revised manuscript provides a clearer, more limited, and more defensible contribution, and we thank you for considering it for further review.

Yours sincerely,

Dr. Stéphane Gaël R. Ekodeck
Corresponding Author
University of Yaounde I, LIRIMA,
Cameroon